Refbank: an open repository of iterated reference game data

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Abstract

Language users flexibly produce and interpret language in context, and over time modulate their language use to convey meanings with greater accuracy and efficiency. Adaptive language use is often studied using iterated reference games, which represent a controlled way of studying how people communicate in the absence of codified labels, and how conventionalized labels form over repeated reference to the same targets. While iterated reference games are a common paradigm, the research questions, pre-processing, and analytic approaches differ across papers so their results cannot be synthesized using meta-analysis. Here, we present Refbank, a harmonized repository of iterated reference game data from 19 different papers, enabling quantitative synthesis of such data in novel and larger-scale analyses. Using Refbank, we conduct five mega-analyses that shed light on the robustness and variability in convention formation under different experimental settings, including analyses of the functional form and moderation of utterance lengths over rounds, and the semantic and syntactic trajectories of descriptions over rounds. Refbank also has potential future uses for targeted experiment design, computational modeling, and language model evaluation. These directions demonstrate the utility of reusing data from iterated reference games, and we invite contributions of additional datasets to Refbank.

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Language is an extensible communication system that can be used to communicate about novel referents, even without established terms for reference. Language users flexibly produce and interpret language in context, and over time modulate their language use to convey meanings with greater accuracy and efficiency (Zaslavsky et al., 2018). Such language dynamics can be studied both at long time scales (e.g., historical linguistics), or at shorter time scales (e.g., discourse analysis), in order to understand how language users adapt to variation and change.

Adaptive language use is often studied using iterated reference games, which represent a controlled way of studying how people communicate in the absence of codified labels, and how conventionalized labels form over repeated reference to the same targets. A *reference game* consists of a group of people communicating with each other about a set of potential referents. One person, the *describer*, has a specific referent, the target, (i.e. an image pre-identified as the target by the experimenter) that they seek to communicate to the *matcher* (or matchers). This act of reference requires coordination between describer and matcher on a symbol–meaning alignment. *Iterated* reference games involve repeated reference to the same targets within the same group of people, so that a certain conversational history builds up that could shape the construction and understanding of later symbols.

Iterated reference games were introduced by Krauss and colleagues in the 1960s (Krauss & Bricker, 1967; Krauss & Weinheimer, 1964, 1966) and popularized by Clark and colleagues in the 1980s (Clark & Wilkes-Gibbs, 1986). This paradigm has been used to study a wide variety of communicative and linguistic phenomena. The primary areas of study are coordination and the formation of conventions or conceptual pacts (Brennan & Clark, 1996; Dahan, 2025; Eliav et al., 2023; Ghaleb et al., 2024; Hawkins, Frank, et al., 2020; Metzing & Brennan, 2003, among others). Some studies focus particularly on iterated reference games as a way of studying collaboration in natural conversations (Clark & Wilkes-Gibbs, 1986; Dahan, 2024; Schober & Clark, 1989; Wilkes-Gibbs & Clark, 1992). Other studies focus on other aspects of coordination, including the negotiation of referring expressions (Bangerter et al., 2020; Knutsen & Le Bigot, 2018) and the role of feedback (Fox Tree & Clark, 2013; Krauss & Glucksberg, 1977; Krauss & Weinheimer, 1966). Iterated reference games with multiple partners have been used to study how speakers design their utterances for their specific partners (Horton & Gerrig, 2002, 2005; Turner & Knutsen, 2021; Yoon & Brown-Schmidt, 2014, 2019a, 2019b) and how speakers might generalize patterns of social conventions across interlocutors (Hawkins, Goodman, et al., 2020; Hawkins et al., 2021). While a majority of iterated reference game studies have focused on typical adult populations, they have also been used with children and teenagers (Branigan et al., 2016; Leung et al., 2024; Turner & Knutsen, 2021) and with older adults (Horton & Spieler, 2007; Hupet et al., 1993; Lysander & Horton, 2012) to study cognitive and linguistic abilities across the lifespan.

A few common patterns recur across papers – participants tend to have high and increasing performance at selecting the correct targets, even as the amount of language produced decreases over repetition. There are also qualitative shifts in the type of language produced, as the descriptions seem to become more conventionalized, sometimes indexed as a transition to more abstract descriptions, and more definite and bare nouns (Brown-Schmidt & Duff, 2016; Clark & Wilkes-Gibbs, 1986; Dahan, 2025; Isaacs & Clark, 1987).

As reviewed above, different researchers have used iterated reference games with different setups and settings as a measure of communication. The breadth of studies in this domain presents an opportunity for evidence synthesis across different samples and experimental designs. In particular, there are a number of dimensions of variation in experimental design across such games, including different stimuli choices, different numbers of options, and different sizes and structures of groups. Running a new experiment across all these options would be expensive, so aggregating existing data provides a way to increase what we can learn from the data we already have, synthesize theories and hypotheses, and design more pinpointed studies in the future (Koile & Cristia, 2021).

Compared to common alternatives, raw data aggregation is an especially useful method for iterated reference games. Systematic reviews are one possible method for evidence synthesis, and can be helpful to map the conceptual and methodological space (Bovet et al., 2023); however, they do not offer quantitative aggregation of results, which is critical in psycholinguistic research since qualitative predictions are often similar across theories. Another common approach for evidence synthesis is meta-analysis; unfortunately, the iterated reference game literature is not ideal for meta-analysis because studies report results for different questions and use different measures for the same questions. Additionally, language data of the kind generated in iterated reference games are very rich and amenable to many different possible analyses, which may not be captured by aggregation into effect sizes, a necessary step in meta-analysis.

In contrast, alignment of raw data in a shared framework (sometimes called "mega-analysis") enables the reuse of high value data by putting it all in one place in a common format and making it easily accessible. The mega-analytic approach has been used to pool data of various types in the language acquisition space, including transcripts of parent–child interactions (CHILDES, MacWhinney, 2000), parent reports of early vocabulary (wordbank, Frank et al., 2017), and data from the looking-while-listening paradigm (peekbank, Zettersten et al., 2022). Similar approaches are used in other domains such as in the OpenNeuro repository for neuroscience (Markiewicz et al., 2021).

One advantage of harmonized dataset repositories is that new analytic approaches can be used on the data, including approaches that may not have been available when some data was originally collected (e.g., semantic vector embeddings), or approaches that rely on cross-study data (Frank et al., 2021). Furthermore, many of the images used in iterated reference games appear in the Kilogram dataset (Ji et al., 2022) which contains whole-image and parts-level descriptions for over a thousand tangram images. Thus, a harmonized dataset of reference game data can also be connected with this other resource to enable image-specific predictions.

There are some key questions about iterated reference games that are best answered with mega-analysis. While datasets consistently show increasing efficiency in picking out a target over rounds (indexed by number of conversational turns, total number of words, words from the describer, or length of describer's first utterance, among other measures), there are open questions around the shape of this reduction curve, which Bögels et al. (2026) suggests is a power law on the basis of different measures from 2 datasets. Checking these patterns across more datasets is important to constraining mechanistic explanations of reduction. Another question is how different experimental factors, including experiment modality and the design of the communication channel, affect the process of convention formation, which

is only testable when there is variation in these factors. Finally, some recent datasets have used NLP tools to look at changes in semantics and part of speech over the convergence process, and mega-analysis can extend these analyses across a broader set of datasets, to look for generalizability across languages, stimuli sets, and modalities.

To enable the reuse of iterated reference game datasets, we present Refbank, an open repository of iterated reference game data. Refbank currently has data from 20 datasets and 8,094 participants (Table 2). In the following sections, we explain the schema of Refbank and what information it contains and describe how researchers can access the data. We then use Refbank to address 5 analytic questions about the generalizability and moderation of iterated reference game phenomena, extending prior studies and demonstrating novel results enabled by mega-analysis.

Design and technical approach

Refbank was constructed to align with the FAIR principles for domain-specific data repositories: findability, accessibility, interoperability, and reusability (Wilkinson et al., 2016). Refbank data is stored on Redivis, a platform for hosting scientific data. A Redivis account can be made freely and is used to access the data. Data is versioned, so that either the current version or a previous version of data can be used (for reproducibility).

Refbank is structured as a relational database with different tables encoding different types of data. Because iterated reference games can vary from each other in different ways, we needed a schema that could accommodate different set-ups while still unifying the information for cross-dataset comparison.

A first question is how to define an iterated reference game. For Refbank purposes, we consider reference games using natural language to communicate (i.e., games where participants provide linguistic expressions to name stimuli so other participants can select them from a group) where at least some of the stimuli are repeated (in at least some conditions), and change across repetition is of interest.

Core tables

The core information in Refbank is stored in three tables: Trials, Messages, and Selections. The key unit in our schema is the trial, which we define as the actions associated with picking out one target image out of a pool of options. Each trial can be associated with messages (spoken utterances or typed messages from the describer and matchers) and selections (which matcher selected which image option).

Trials. Each row in the Trials table contains information about a single trial including what was happening that trial and how it fits into the larger structure of a game.

Each trial is associated with a describer and a list of matchers via a set of unique identifiers (player IDs). Each trial has a target image (seen by the describer) and a set of image options (that matchers choose among), linked via their image IDs. We also record the option set size – how many images were possibilities that trial.

Each trial is labelled as either an "order" trial or a "match" trial. In matching trials, a given target is specifically highlighted and the matcher picks one and then moves on to the next trial. In ordering, the goal is to reproduce a given order of the set (or subset) of targets. We separate these into one trial per target (e.g., ordering 12 tangrams would correspond to

12 trials), but there is flexibility in that a matcher might revise their choice to one trial after later trials within the same round.

The basic structure of an iterated reference game is a series of trials (acts of picking out different targets) that are nested within rounds (sometimes called blocks or repetitions), where each round typically involves one trial per target in the target set. In a small number of cases, the targets for a round were sampled from a larger target set, and thus participants did not see all possible targets each round (e.g., if there are 12 targets in total, but participants only saw 3 per round).

The Trials table also contains information about the games that trials are nested within. Each game is a set of people who play with each other, possibly on a network of interlocking pairs or with substitutions (see metadata). Following Hawkins et al. (2021), we include a room number for when there are two trials going on in parallel in the same game (this occurs on network games where two pairs play and then swap).

If there is a change in how the game is working, such as a swap in partners, or in the pool of stimuli, we define that as a new stage, recorded via the stage number. For example, some experiments (Hawkins et al., 2021; Hawkins et al., 2023) have groups of 4 people play with each other in a round-robin design, where people play with one partner for a few rounds and then swap partners. In these cases, each set of pairings is a stage, and the first stage of data can be treated as coming from pairs (since no partner switches have yet occurred). The stage designation is also used for experiment designs with entrainment and test phases (Yoon & Brown-Schmidt, 2019a) and experiments that have a separate "generalization" round at the end with new stimuli (Ji et al., 2025). In general, restricting to only "stage 1 data" (as is possible in the visualization app and via filtering downloaded data) allows for a more even comparison across games.

While the trials table includes information about the setup of the trial, the actual content of the trial consists of two kinds of actions: messages and selections.

Messages. Each message is one turn of dialogue or one message that was typed in a chat window, depending on the game modality. A message has text, the content of the message, and is associated with a player ID, who produced the message. Messages are also associated with the player's role (either matcher or describer). All messages have a trial ID (linking to the trials table). Off-topic messages such as chit chat or meta-discussion about strategy are included, but marked via a message irrelevant column, to allow for filtering. There is also a time stamp field for when the timing of a message within a trial is known. We do not remove off-topic messages, or trim messages with some extraneous content, but we do use spelling-corrected versions of written messages where available.

Selections. Players' selections are recorded in a selections table. Each selection is tied to a given trial and a given player ID. Each selection is associated with a selected image that cross references the image id in the images table (and can be compared with the target image for the trial to calculate if the selection was correct). There is again the option for a time stamp to specify how long into the trial the selection was made.

In order games, we only record the final selections players made for each position.

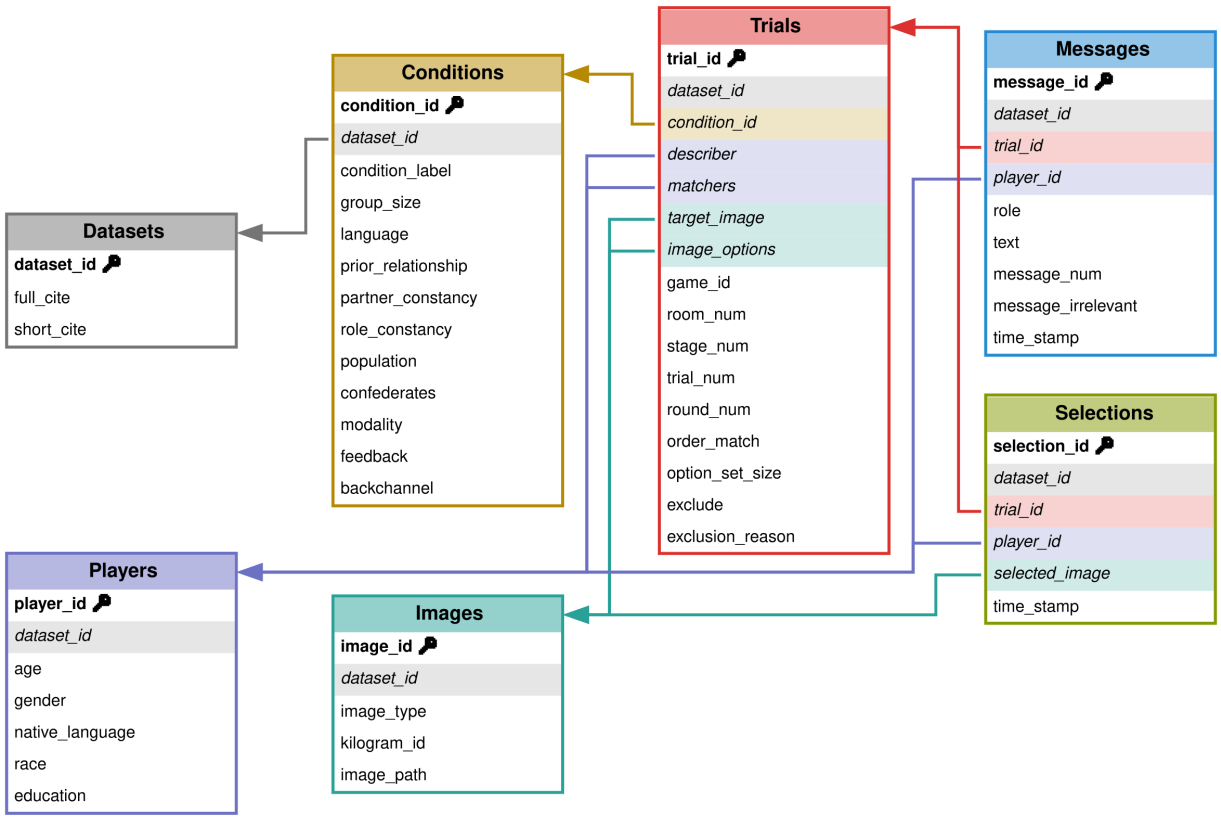


Figure 1. Diagram of the core tables in the schema and how they connect to each other.

Metadata

In addition to the core information about what happened during the game, we have several tables for metadata about the players, images, and experimental features of the study.

Datasets. The datasets table contains only bibliographic information, linking our dataset ID to the full and short citations for that dataset.

Conditions. Iterated reference games can vary along a number of design features. Most of these factors are coded in the "conditions" table. We code several metadata factors as condition information. These include:

- language: the language in which the iterated reference game was conducted
- group-size: the number of players in a game, where we define game as a set of players who interact with each other but no one else
- partner constancy: whether or not the same set of people is active the entire game. If there are changes in who is interacting, there is not partner constancy.
- role constancy: roles are constant if each player is only the describer or a matcher when they are active; roles are not constant if a matcher becomes a describer or vice versa.
- backchannel: designs vary on how the matchers may communicate with the describer. We break this up into 3 levels: full if there are no restrictions on communication (within the modality), limited if there is some communication (example, if only one message can be sent), and none if there is no communication channel from matcher to describer.

Table 1
Design features of each dataset. 'ltd' is used as an abbreviation of limited. Multiple values indicate that the paper reported on different conditions that varied in the given feature.

Paper	Language	Modality	Role constant	Partner constant	Feedback	Backchannel
Bangerter & Smolenski (2000)	Swiss German	oral-in-person	no	yes	none	full
Bangerter et al. (2020)	French	oral-in-person	yes	yes, no	none	full
Beatty-Martinez et al. (2026)	Spanish/English	written	no	yes	full	full
Bögels et al. (2026)	Dutch	oral-in-person	no	yes	none	full
Boyce et al. (2024)	English	written	mixed	yes	ltd, full	ltd, full
Boyce et al. (2026)	English	oral-in-person	no	yes	full	full
Branigan et al. (2016)	English	oral-in-person	yes	no	ltd	full
Dahan (2023)	English	oral-in-person	no	yes	none	full
Dahan (2025)	English	oral-in-person	no	yes	none	full
Dale et al. (2011)	English	oral-remote	yes	yes	none	full
Hawkins et al. (2020)	English	written	yes	yes	full	none
Hawkins et al. (2020)	English	written	yes	yes	full, ltd	full
Hawkins et al. (2021)	English	written	no	no	full	full
Hawkins et al. (2023)	English	written	no	no	full	full
Hawkins et al. (2026)	English	oral-remote	no	yes	full	full
Ji et al. (2025)	English	written	no	yes	ltd, full	none, full
Leung et al. (2025)	English	oral-in-person	no	yes	none	full
Mankewitz & Hawkins (2025)	English	written	no	yes	full	full
Wang et al. (2025)	English	oral-remote	yes	yes	none	full
Yoon & Brown-Schmidt (2019)	English	oral-in-person	yes	no	none	full

- feedback: we use "feedback" to refer to information players receive on the correctness of their selections from the experimenters. In full feedback, after each trial, players learn if the selection(s) were correct or incorrect and what the correct target was. In limited feedback, there is some correctness feedback, but it doesn't include what the correct target was, or is provided after every round, perhaps as a total correct. In no feedback, players receive no feedback during the game.
- modality: We code the communication modality into three levels: oral-in-person when the participants are talking and physically in the same room, whether or not they have visual access to each other; oral-remote when the participants are talking over some sort of connection (audio signal between phone booths, headsets in separate fMRI machines, etc). We use written when the messages are typed, like via a chatbox.
- confederates: we include an option for noting if one or more of the participants was not a "real" participant – i.e., was an experimenter or an AI bot. So far, this is relevant to only one study in Refbank, but as iterated reference games involving some artificial intelligence agents are a topic of recent study, we anticipate more might be included in the future (Hua & Artzi, 2024; Jones et al., 2026; P.-Y. A. Wang et al., 2026; Zeng, Li, et al., 2026; Zeng, Paige, et al., 2026). By marking them with a metadata flag, these games can easily be treated separately or excluded as appropriate for the research goals. Games with confederates are excluded from the visualization app.
- prior-relationship: another possible source of design variation is whether the pair or group of participants have prior history with each other, which could shape their interactions and convention formation. In some cases, the populations were mixed,

either with an explicit balance between familiar pairs or stranger pairs, or because the recruitment pool included some classmates and some strangers.

Players. The player table contains some basic demographic information about participants when available. It can be linked via the player IDs. We track players' age, gender, native language, race, and level of education, although these are not always available across datasets.

Images. Each image has a unique image ID for cross referencing with the trials and selections tables. We also record image type (tangram, line drawing, fribble, etc). For tangram images that are in the Kilogram dataset (Ji et al., 2022), we include the Kilogram ID for cross-referencing. When we have access to the actual image file, we store these and include their image path. The same images are often reused across different datasets, and the image records are deduplicated so multiple papers may refer to the same image ID.

Derived

The core and metadata tables represent the raw data that we import from individual studies. We also have a workflow for processing the raw data and creating derivative measures that may be commonly of use to researchers. These derivative tables are created via a Redivis workflow. For example, word counts can be derived from the text data in messages but are a common enough metric that we calculate them for a summary table. Other derived tables contain processed representations of the describer's messages, including semantic vector embeddings (using Jina AI embeddings, Akram et al., 2026) and dependency parses and part of speech tags (using Stanza, Qi et al., 2020). In producing these measures, we apply our own light exclusions, requiring that the describer did produce descriptions, and that there are at least 2 rounds worth of trials for that game. As additional processing pipelines are developed, we may add more derived tables.

Dataset processing and import

We started developing Refbank with the iterated reference games that were conducted by members of the team, or where the data was already posted publicly. We then contacted authors of a number of iterated reference game papers asking if they were willing to share their data.

Rather than hand create a linked set of tables in the Refbank format, we took raw datasets and reformatted them into one wide table with one row per choice or message in individual R import scripts. We then ran a validation and import script that checked the columns and values and split the wide table into the linked tables for use with redivis. We have another script that reruns all dataset import scripts, does more validation, creates unique identifiers across datasets, and upload the tables to redivis. This script is run whenever we release a new version of Refbank (for instance when a new dataset is added). We checked datasets against the published results using a script that applied the exclusions and produced graphs and summary statistics; these per-dataset summaries are available on the datasets tab of the Refbank website for external checking.

We received data in a variety of formats, including pre-cleaned tabular data, output from experimental software like Empirica (Almaatouq et al., 2021), transcripts, and raw audio files. Datasets varied in how much effort they took to import and align to the Refbank

schema. The easiest datasets already had trial-level data in an easy to parse format (csv or json) where we only had to reshape the data and rename the columns to match the Refbank import schema. This was the case for many of the more recent datasets with online data collection and typed descriptions.

Datasets were more difficult to import if there were file parsing complications (conversion from .doc files), if the dataset needed to be transcribed, and if we had to infer trial boundaries, speaker identities, and target images. These more difficult imports were often from spoken datasets that are more difficult to collect and thus have higher re-use value.

We transcribed raw audio for 2 datasets (Dahan, 2023; Hawkins, 2026) using WhisperX (large-v3, Bain et al., 2023). When transcripts or transcribed audio did not have trial boundaries, we hired a contractor to annotate trial boundaries by hand (and confirmed high agreement with an author) or annotated trial boundaries by hand [5 datasets; Bangerter and Smolenski (2000); Bangerter et al. (2020); Dahan (2023); Hawkins (2026); Hawkins, Frank, et al. (2020)] . When we had the pool of target images but did not know which image went with which trial (or target label), we marked the correspondences we thought were most likely (2 datasets, Bangerter & Smolenski, 2000; Branigan et al., 2016) . Each dataset has a README including citation information, the abstract, the key facts about the study design, and notes about any inferences or processing we had to do as part of the import process. These README's are viewable on the "Datasets" tab of the refbank website, in addition to in the repository with import scripts.

Datasets also varied in completeness; some datasets had timing information and demographics, whereas in others only the transcripts were available, or only selection data. We have imported what was available, but due to incomplete data, some datasets might not be applicable for all analyses.

We imported trials from all games, including games that were excluded in the initial studies. We mark their exclusion status and reason, but allow users of Refbank to impose their own uniform exclusion criteria.

Metadata was coded from the data and from the paper or write-up. In a few cases, condition information was not fully specified in the paper. In these cases, we either talked with the authors, or made our best guesses from the paper. When there were borderline cases that were not explicit we inferred them and made a note in the README.

Some of the contributed datasets had additional information beyond the shared metrics. We do not include these additional data (such as pre- or post- tests, mouse or eye tracking, fMRI data, or audio files).

Current datasets

Refbank currently has data from 20 datasets including 1,879,919 words from 149,654 referential trials from 2,774 games and 8,094 participants (Table 2).

These datasets are primarily in English, although Refbank includes one dataset each in French, Swiss German, and Dutch, and one dataset with codeswitching between Spanish and English. Datasets are primarily made up of adult participants, although 3 datasets include data from children: two of just children interacting with peers (Boyce et al., 2026; Branigan et al., 2016), and one of child–parent pairs (Leung et al., 2024). Datasets vary in their design

features (Table 1). These dataset characteristics reflect both the distribution of papers published and what datasets were made available to Refbank.

We welcome any additional datasets, including unpublished datasets. Transcripts, selections, and experimental metadata are all de-identified and so do not count as "human subjects data" under United States regulations, and so likely can be publicly shared.

Table 2

Amount of data from each dataset, and the amount of data per game in each dataset. Ranges indicate that the value varied across conditions within the dataset. Option set sizes in parentheses indicate that a minority of trials has this other option set size.

Paper	Total games	Total players	Total trials	Total words	Trials/ game	Rounds/ game	Group size	Option set size
Yoon & Brown-Schmidt (2019)	788	3 334	41 766	348 849	80 - 208	5 - 13	4 - 7	16 (4)
Mankewitz & Hawkins (2025)	495	993	31 298	220 551	64	4	2	4
Boyce et al. (2024)	306	1 255	21 269	374 105	72	6	2 - 6	12
Ji et al. (2025)	262	524	8 183	50 468	30 - 40	5 - 6	2	10
Hawkins et al. (2020)	173	346	12 376	128 274	24	6	2	4
Hawkins et al. (2020)	173	346	12 376	128 274	24	6	2	12
Hawkins et al. (2020)	173	346	12 376	128 274	72	6	2	4
Hawkins et al. (2020)	173	346	12 376	128 274	72	6	2	12
Hawkins et al. (2020)	168	279	3 994	19 740	24	6	2	4
Hawkins et al. (2020)	168	279	3 994	19 740	24	6	2	12
Hawkins et al. (2020)	168	279	3 994	19 740	72	6	2	4
Hawkins et al. (2020)	168	279	3 994	19 740	72	6	2	12
Boyce et al. (2026)	102	199	1 513	18 018	12 - 16	3 - 4	2	2
Leung et al. (2025)	82	165	3 280	47 909	40	4	2	2
Bangerter et al. (2020)	76	182	3 757	161 149	40 - 64	5 - 8	2 - 3	8
Bögels et al. (2026)	47	95	4 512	176 483	96	6	2	16
Beatty-Martinez et al. (2026)	45	91	2 698	30 848	60	5	2	12
Hawkins et al. (2021)	43	172	3 699	25 777	96	12	4	4
Hawkins et al. (2026)	42	86	3 761	96 102	180	8	2	18 (36)
Hawkins et al. (2023)	26	105	2 278	11 756	96	12	4	4
Branigan et al. (2016)	24	74	764	30 430	32	4	3	8
Dahan (2023)	20	40	960	49 071	48	16	2	16
Dahan (2025)	20	40	960	47 297	48	16	2	16
Dale et al. (2011)	20	40	359	0	18	3	2	6
Wang et al. (2025)	20	40	800	26 105	40	4	2	13
Bangerter & Smolenski (2000)	15	34	1 427	16 987	96	12	2	8
Total	2 774	8 094	149 654	1 879 919				

Using Refbank

There are several ways of interacting with the Refbank data. We recommend using the Observable web application for quick visualizations and exploration of moderators or different datasets, via the Refbank website (refbank.stanford.edu). The Refbank website can also be used to learn about the source and characteristics of each dataset and compare a summary of the refbank data against the original report. For analyzing data, the data tables can be downloaded via the Redivis API for Python, or using `refbankr`, a custom R package. `refbankr` functions can be used to pull any set of datasets at any version, and also to pull a sample of the data for checks.

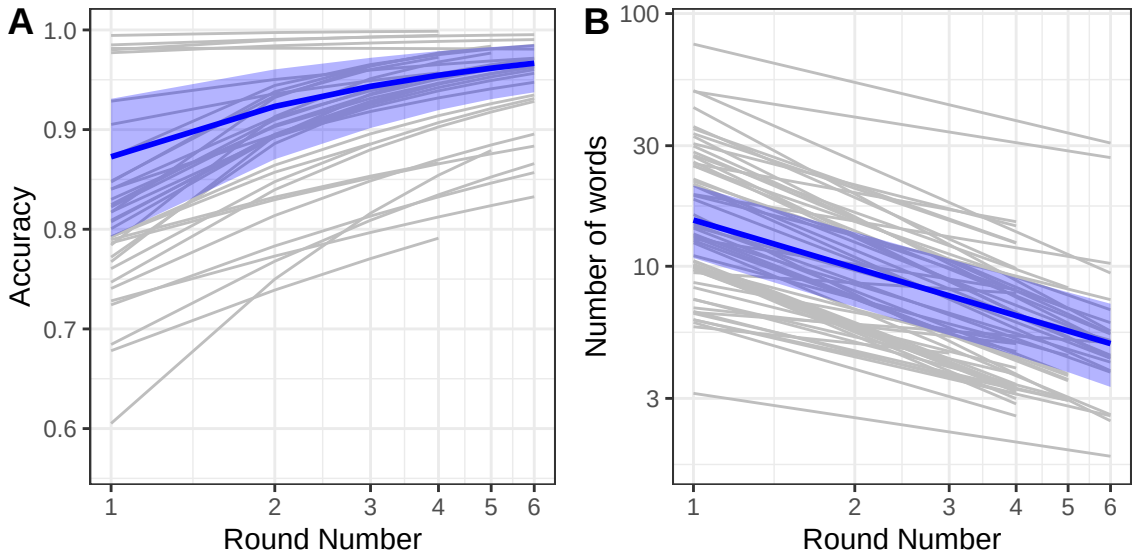


Figure 2. Model predictors for accuracy (A) and describer utterance length (B). Each condition is shown in a faint line, and the generalized prediction is shown in blue, with a 95% CI.

Redivis has built-in methods of filtering, exploring, and downloading data, via its viewers and workflow creation, where a chain of SQL transforms, and R and Python notebooks can be used to process data. Redivis allows for the creation of workflows that pull from datasets, and for SQL-type queries over datasets.

Different workflows support different possible uses including scientific research and training. The API format means that Refbank data can be pulled as part of larger pipelines and processing involving coding agents.

Analyses

Refbank enables a range of analyses that are only possible with larger and more diverse datasets. Here, we present five mega-analyses that shed light on the robustness and variability in convention formation under different experimental settings. We find that (1) participants are very accurate and improve rapidly over rounds, (2) participants use fewer words over rounds, with a shape best described by a log–log function, (3) this reduction is moderated by participant age group, backchanneling, and option set size, (4) descriptions robustly converge semantically within games and diverge across games, and (5) modality moderates the change in part of speech proportions across rounds.

For these analyses, we only analyzed games and targets where the target was referred to at least 3 times during the game and where message data was available, and we only analyze data from the first stage of the game, before any swaps of stimuli sets or changes in partners.

Accuracy

A common finding across iterated reference games is that players are usually quite accurate at communicating and selecting the correct targets – that is, they succeed at

reference. When participants are not at ceiling initially, they tend to improve rapidly over rounds. A high level of accuracy is taken as a pre-condition for interpreting the language data as being a sign of conventionalization.

Not all datasets included accuracy data on the trial level, so these results come from the 15 datasets (136,643 trials) where selection data and text data were available.

To aggregate across datasets, we ran a mixed-effect logistic regression model correct $\sim$ log round num $\times$ option set size + (log round num||dataset/condition) to model the correctness of each selection from each matcher. For computational tractability, we do not include per-group or per-individual random effects and we use uncorrelated random slopes and random intercepts. We include the number of options as a predictor since this affects the baseline chance of correct selections.

The fitted predictions are shown in Figure 2A. The overall fit line is shown for a 12 image option set. Consistent with previous findings, accuracy is generally high to start with, but also increases over rounds (odds ratio of one log-round later 2.08 [1.44, 2.94]).

Reduction of utterance length

A very consistent finding across many iterated reference games is that the amount of language used in descriptions declines across rounds. However, while the trend is frequently observed, the specific mechanisms are not known, and the functional relationship between round and quantity of words has not been firmly established. Bögels et al. (2026) compared a linear model and a log-log (power law) model on two datasets (Eijk et al., 2022; Hawkins, Frank, et al., 2020), and found that log-log model provided a better fit to the data.

While there are many possible functional relationships between repetition and number of words, here we consider four options in order to determine what general type of shape fits the data best. By using a larger number of datasets that vary in different ways, we avoid overfitting to the peculiarities of any one dataset.

We consider relationships between round and words on either the raw or log scales for each. For each model we included mixed effects (e.g., log words $\sim$ round num + (round num||dataset/condition)). We found that the best fitting model was log(words) $\sim$ log(round), which was a better than log(words) $\sim$ round (ELPD difference of -1157, SE = 46). Models with a linear number of words were much worse (ELPD differences of -97,884 and -100,179).

Using the best fitting log-log model, we confirm the common finding that there is a substantial decrease in the number of words over rounds (Figure 2B). There was a change of -0.63 [-0.75, -0.51] log words per log round number.

Reduction moderators

One analysis enabled by the aggregation of data across studies is the examination of the possible effects of experiment design features on the rate of reduction – that is, whether some conditions result in faster reduction than others. For this analysis, we used the log(words) $\sim$ log(round) model and added moderators as predictors of slope change. We used the moderators coded in the conditions table, but collapsed some of the 3-value moderators into binary distinctions. The moderators we considered were population (only adults vs children or child/parent), backchannel (full vs limited or none), feedback (full vs limited or none), modality (oral vs written), role constancy, number of image options, and number of

players. Number of image options and number of players were log transformed. Partner constancy is listed in the conditions table, but this analysis was run only on stage 1 data before any partner changes occurred, so we did not include it as a predictor. We did not include interactions between predictors due to the sparsity of some moderator combinations. We also note that existing datasets do not sample uniformly across the moderator space, and some predictors were correlated in their distribution across datasets (Figure 3A).

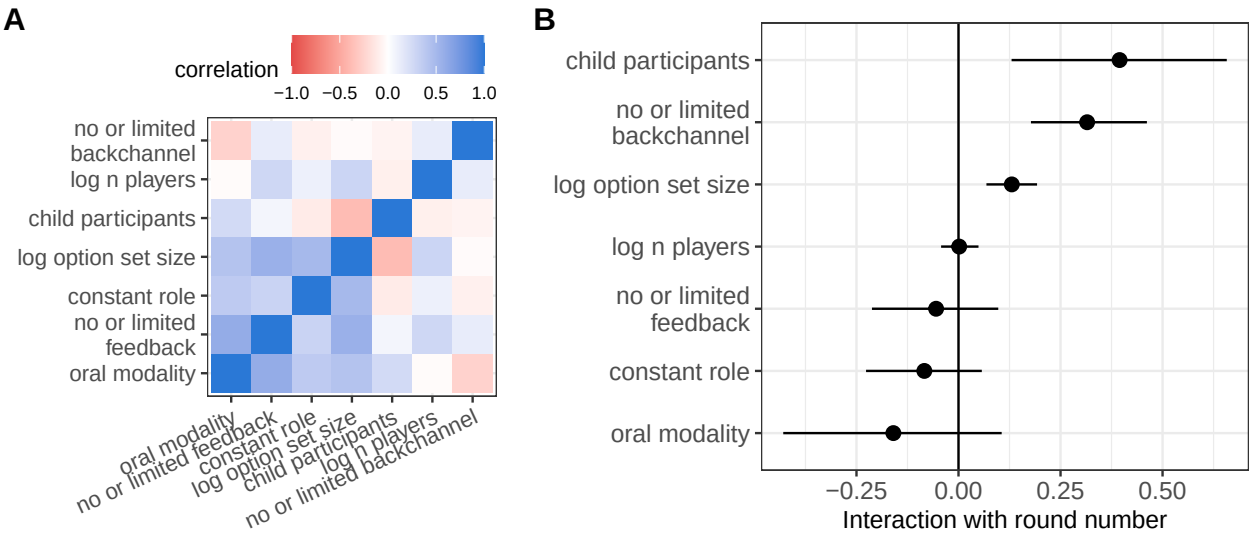


Figure 3. Panel A: Correlations between moderators. Panel B: Coefficients for different predictors interacting with log round in a log-log reduction model. Point estimates and 95% CrI are shown. Negative values indicate faster reduction.

Figure 3B shows the fitted coefficients for the different predictors. Games with children, games with no or limited backchannel, and games with larger numbers of images had slower reduction in the number of words. Group size had no effect on reduction. Limited or no feedback, constant role, and spoken games all had point estimates of faster reduction, with uncertainty over direction and magnitude of the effect.

Given the correlations between some of the predictors, and the absence of interaction terms from our model, we cannot interpret these effects as causal, merely predictive in these datasets. For example, less feedback and oral modality are positively correlated, which could explain the counter-intuitive (weak) association between less feedback and faster reduction. Presence of a backchannel might also have interaction with modality, if backchanneling has different affordances in spoken and written interactions, but we do not include the interaction, because we do not have any datasets with oral modality and restricted backchannel.

Semantic embedding similarities

One way of quantifying the semantic trajectories of descriptions over rounds is to embed the descriptions in a semantic vector space and then quantify similarities of pairs of descriptions using cosine similarity or another distance metric. This approach is used in analyses of dialogue trajectories (Fischer & Ram, 2024; Reece et al., 2023).

As part of the Refbank processing pipeline, we embed all descriptions using a multilingual text embedding model, specifically jina-embeddings-v5-text-small (Akram et al., 2026; Feng et al., 2022). We then calculate cosine similarities between relevant pairs of descriptions to measure how the semantic distances of descriptions change across round number.

Within a game, the similarity between descriptions to the same target in adjacent rounds is an index of convention formation (Boyce et al., 2024; Hawkins, Frank, et al., 2020). When there is low similarity, descriptions are changing, but when there is high similarity the target to description mapping has stabilized. To test for stabilization of descriptions, we ran a mixed effects ordered beta regression model sim $\sim$ log round num + (log round num||dataset/condition) (Kubinec, 2023). In this model, the similarity between a description and the corresponding description in the next round increases over rounds by 0.83 [0.55, 1.08] per log round. The predicted trajectories for a generic reference game and each individual condition are shown in Figure 4A.

Similarity metrics can be affected by sentence length, so as a robustness check, we ran a model controlling for the length of both descriptions. Controlling for description lengths, the increase in similarity between adjacent rounds was smaller, but still substantial (0.49 [0.27, 0.69]).

As games converge to a variety of different, arbitrary conventionalized names, these descriptions often become differentiated from one another (Boyce et al., 2024; Hawkins, Frank, et al., 2020). To test for divergence between different games, we ran a mixed effects ordered beta regression model sim $\sim$ log round num + (log round num||dataset/condition) predicting the average distance of a description to other descriptions for the same round and target image, but a different game in the same condition. We included averaged similarities when there were at least 4 descriptions to compare to. In this model, the similarity between descriptions from different games changes by -0.19 [-0.25, -0.13] per log round. The predicted trajectories for a generic reference game and each individual condition are shown in Figure 4B. Controlling for description lengths, the decrease in similarity across games was smaller, but still substantial (-0.12 [-0.16, -0.07]).

Distribution of parts of speech

The distributions of words and phrases used in descriptions change as conventionalization occurs. For instance, definite or bare noun phrases may increase compared to indefinites (Brown-Schmidt & Duff, 2016; Dahan, 2025; Isaacs & Clark, 1987), and more holistic or metaphorical descriptions may become more common relative to literal, shape-based, or part-based descriptions (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986). As an illustration of how Refbank and natural language processing pipelines can be used to scale up reference game analyses, we focus on the change in the distribution of parts of speech. Hawkins, Frank, et al. (2020) found that the distribution of parts of speech in describer utterances changes across repetitions, with nouns becoming relatively more common at the expense of closed-class words. We are able to scale up this analysis to all the monolingual datasets in Refbank. We use Stanza to parse each message (Qi et al., 2020). The part of speech distributions and full dependency parses are saved in derived tables in Refbank.

For this analysis, we lump parts of speech into 6 coarse grained categories: nouns,

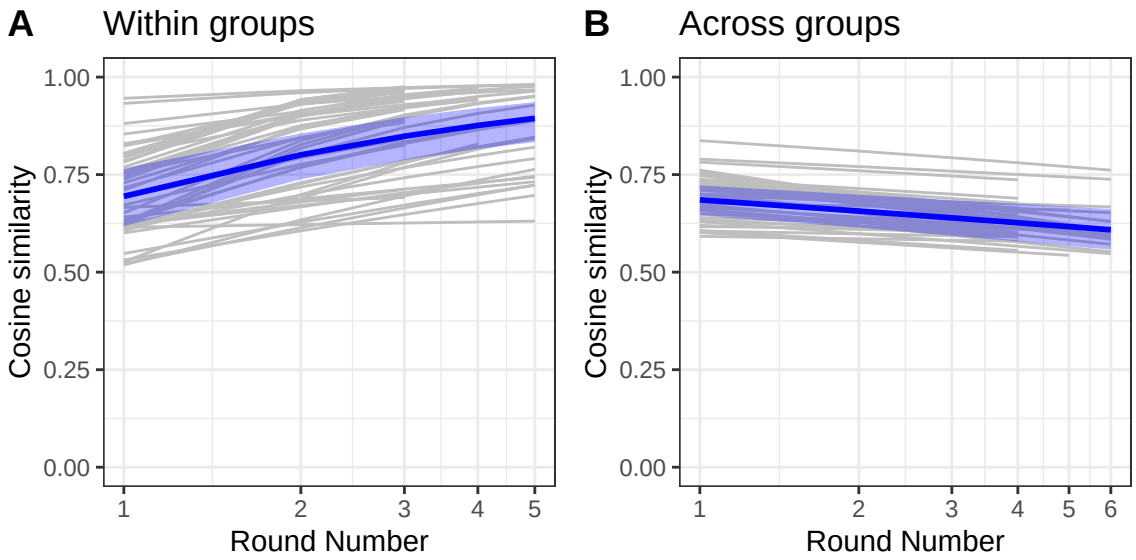


Figure 4. Cosine similarities between a description and the next round description from the same game (A) or descriptions from the same round and different games (B). Blue line shows the overall model prediction and 95% CrI, faint lines show per-condition predictions.

verbs, modifiers (adjectives and adverbs), pronouns, determiners, and function words (adpositions, conjunctions, etc). We ran a mixed effects multinomial model with pronoun as the reference class, weighting each trial equally: cbind(noun,verb,modifier,function,det,pron)|trials(total) + weights(w) ∼ log round num × modality + (log round num||dataset/condition). After initial visualization suggested that spoken and written corpora might vary, we added modality (oral or written) as a predictor in an exploratory model. According to the model predictions, from the first to sixth round, the fraction of nouns in describer utterances is expected to increase 0.16 [0.05, 0.27], while function words (-0.07 [-0.12, -0.01]) and pronouns decrease in frequency (-0.05 [-0.06, -0.04]). Other categories were less consistent. Modality makes the largest difference in the size of the noun boost (0.11 [0.02, 0.21] for spoken versus 0.20 [0.07, 0.32] for written) and the direction of change for determiners, where determiner rate is neutral to increasing in spoken corpora (0.05 [-0.02, 0.14]) and decreasing in written corpora (-0.08 [-0.13, -0.04]). The model predicted results from the model with modality are shown in Figure 5.

Discussion

Iterated reference games are a classic paradigm that have been used to study numerous aspects of dialogue and communication, and continue to be used in new ways (e.g., with language models). While iterated reference games share some core properties, games can also vary along a number of design parameters, and analytic pipelines can vary in the data processing used and results reported. Even for research questions that are commonly reported on, such as change in number of words over repetition, differences in design and modeling approach and reporting make it difficult to quantitatively synthesize results. Taken together, these factors mean that iterated reference games are well-suited for a harmonized repository: data can be unified, data have high reuse value, and alternative synthesis

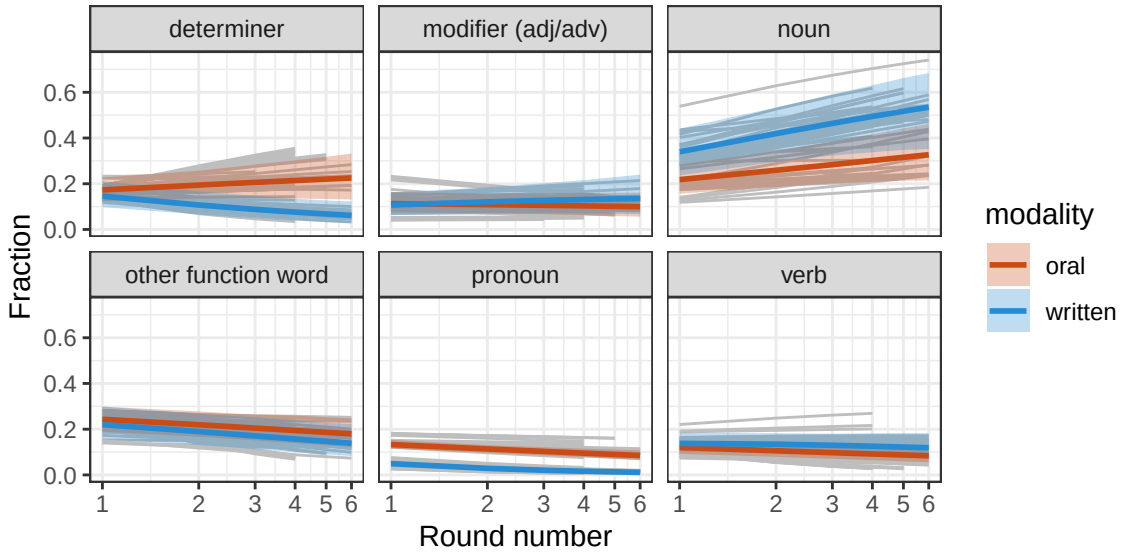


Figure 5. Distribution in parts of speech over rounds. Colored lines and ribbons are the predictions and 95%CrI from the model; faint lines are per-condition model predictions.

methods such as meta-analysis are not as suitable.

In this paper, we introduced Refbank, an open repository of iterated reference game data. We then presented a series of analyses using Refbank data to quantify patterns of generalization and moderation across data from different experiments, contributing to the characterization and understanding of convention formation.

The functional relationship between utterance length and round is more robustly fit over a range of datasets, and we provide additional evidence that the relationship is roughly log–log as suggested by Bögels et al. (2026). Moreover, using the coded metadata, we were able to examine which design features predict the slope of reduction. We found that child participants, limited backchannel, and larger option size sets are associated with slower reduction, although we cannot make causal attributions. Knowing whether certain design features matter is important to experiment design, but requires large amounts and diversity of data to study. This does point the way towards features that should be manipulated in future controlled experiments or included in generative models of reduction phenomena.

Semantic trajectories have been analysed in iterated reference games before, but by scaling up a unified pipeline, we are able to analyse multiple datasets, including in different languages, in the same pipeline. We confirm that the trends of within game semantic convergence and between game semantic divergence extend to spoken studies, studies using other stimuli, and studies in different languages.

Cross-dataset comparisons also allows Refbank to enable novel findings: In an analysis of distributions of part of speech across round, we found some divergences between spoken and typed paradigms, where noun proportion increased more sharply in typed datasets, and determiners increased in spoken datasets but not written ones. This may reflect differences in the cost to produce different parts of speech in different modalities. A more telegraphic approach may be easier in typing where cost is per character, but in speaking, determiners and other short function words may be easier to say than to omit. This idea of differential

cost based on modality suggests a direction for future work that more closely manipulates modality and cost in controlled, comparable experiments.

By unifying datasets in a common framework, we make possible the generalization of existing analyses to a larger collection of datasets, the use of new processing pipelines on the data as techniques arise, and the study of variation across datasets and design choices. Additionally, harmonizing raw data retains individual- and dyad-level patterns, which are important for individual difference analyses, and generative and computational approaches which require raw data to compare against. For example, iterated reference games can be used to assess the extent to which language models are able to communicate like humans, and actual trial-by-trial data are the most useful for such comparisons (**tan2026onceitgetspreprinted?**).

Data aggregation is also a starting point for future experiment design. Refbank data can shed light on what aspects of design may be more or less impactful, and may also allow for the selection of stimuli that have desirable properties. Aggregation can also highlight underexplored regions of the experiment design space, suggesting new avenues for iterated reference game research.

Limitations

While Refbank provides size and diversity compared to any individual study, Refbank's generalizability is limited by the datasets that are collected and the datasets that are shared with Refbank. Some papers that collected reference game data no longer have their raw data or were unwilling to share. Additionally, most iterated reference games are conducted in English, reflected in the linguistic skew of Refbank. A broader distribution of languages is necessary for some questions around how language changes in a reference game.

While Refbank has these limitations, it can also be used to identify gaps in the current pool of reference games and point the way to future high value experiments. The data are heterogenous, but metadata is coded, so researchers can filter to the appropriate datasets, games, and trials for their analyses.

Conclusion

We encourage the sharing of iterated reference game data and hope that Refbank will house more games in the future. Refbank can be used to study the moderators and generalizations of convention formation phenomena in iterated reference games, and identify gaps where future experiments could resolve theoretical questions. Refbank is also a corpus of task-specific dialogues that can be used for corpus linguistics, for assessing pragmatic phenomena in language models, and as a pedagogical resource for instructors and students alike.

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